외계행성 대기
Exoplanetary Atmospheres

Lecture 5: Radiative Transfer III — Albedos, Phase Curves, Line Formation, and Solutions
2026 October 12 (Monday)

Based on: Chapters 3, 5 and 6 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 2 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 5

Planetary Albedos (continued)

Planet-Star Flux Ratios

Planetary Phase Curves

The Emergent Spectrum: A Conceptual Picture

An Introduction to Line Formation

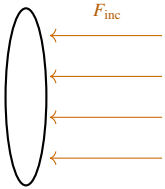
Solutions I: Transmission and Thermal Emission

Solutions II: Scattering Atmospheres

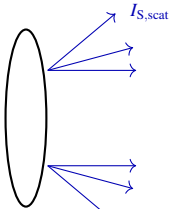
Monte Carlo Radiative Transfer

Summary

The Lambert-disk flux and the meaning of $I_{\rm inc}$



incident flux $F_{\rm inc}$



scattered flux
$F_{\rm scat} = \pi I_{\rm S,scat}$

Scattered flux. A Lambert disk scatters with the same intensity $I_{\rm S,scat}$ into every outgoing direction, so the hemispheric integral of $\cos\theta$ gives $F_{\rm scat} = \pi I_{\rm S,scat}$.

Energy conservation. A perfect Lambert disk absorbs nothing:

$$F_{\rm inc} = F_{\rm scat} = \pi I_{\rm S,scat} \quad\Longrightarrow\quad I_{\rm S,scat} = \frac{1}{\pi} F_{\rm inc} \overset{\text{define}}{\equiv} I_{\rm inc}.$$

Although starlight arrives as a directional beam, $I_{\rm inc}$ is the equivalent intensity that would give the same $F_{\rm inc}$ if the incident radiation were isotropic over the incoming hemisphere: $F_{\rm inc} = \pi I_{\rm inc}$. It also equals the intensity scattered back by a perfect Lambert disk.

Flux at Earth. Seen face-on at distance $D_\oplus$ (in terms of latitude coordinates on the disk),

$$\begin{aligned} F_{\rm L\,disk} &= \left(\frac{R_p}{D_\oplus}\right)^2 \int_{-\pi/2}^{\pi/2}\int_{-\pi/2}^{\pi/2} I_{\rm inc} \cos\phi \cos^2\theta \, {\rm d}\theta \, {\rm d}\phi \\ &= \left(\frac{R_p}{D_\oplus}\right)^2 \pi I_{\rm inc}, \end{aligned}$$

since $\int \cos\phi \, {\rm d}\phi = 2$ and $\int \cos^2\theta \, {\rm d}\theta = \pi/2$.

The geometric albedo: general expression

Recall (Lecture 4, p. 48). The geometric albedo is the ratio of the planet's observed flux at zero phase angle (full phase) to that of a Lambert disk at the same distance and with the same cross-sectional area:

$$A_g(\nu) \equiv \frac{F_\oplus(\nu, \alpha = 0)}{F_{\mathrm{L\,disk}}(\nu)}.$$

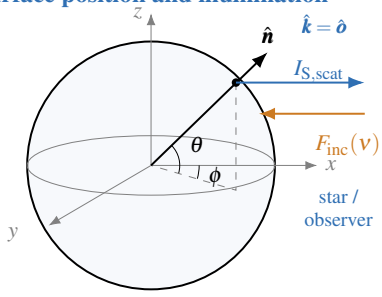
Therefore,

$$\boxed{A_g(\nu) = \frac{\displaystyle\int_{-\pi/2}^{\pi/2}\int_{-\pi/2}^{\pi/2} I_{\mathrm{S,scat}}(\theta, \phi, \nu) \cos\phi \cos^2\theta \, \mathrm{d}\theta \, \mathrm{d}\phi}{\pi I_{\mathrm{inc}}(\nu)}}$$

- Something interesting: the denominator $\pi I_{\mathrm{inc}}(\nu) = F_{\mathrm{inc}}(\nu)$ is the incident stellar flux at the *substellar point*. So A_g may also be described as the ratio of the planet's scattered flux to the incident stellar flux at the substellar point.
- This is the form to use to compute A_g from the output of a *model atmosphere code*, which delivers the scattered intensity at the planetary surface as a function of location, $I_{\mathrm{S,scat}}(\theta, \phi, \nu)$.
- The subscript "scat" means that only *scattered* radiation is counted, not radiation that has been absorbed and thermally reradiated; at visible wavelengths, for planets cooler than ~ 1500 K, the distinction is clean.

Geometric albedo: surface geometry and intensity definitions

Surface position and illumination



θ: latitude $\quad$ ϕ: longitude

$$\mu_i \equiv \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{k}} = \cos\Theta$$

Full phase: $\mu_i = \mu_o = \cos\theta\cos\phi$

Incidence angle Θ is measured from the local normal.
The diagram shows a surface point at full phase.

Flux and equivalent intensity

At the substellar point, define the equivalent isotropic intensity by

$$I_{\mathrm{inc}}(\nu) \equiv \frac{F_{\mathrm{inc}}(\nu)}{\pi}.$$

At a surface element (θ, ϕ):

$$F_{\mathrm{in}}(\theta,\phi,\nu) = \mu_i F_{\mathrm{inc}}(\nu) = \pi \mu_i I_{\mathrm{inc}}(\nu).$$

For a Lambert surface

$$p(\nu) \equiv \frac{F_{\mathrm{scat}}}{F_{\mathrm{in}}}, \qquad F_{\mathrm{scat}} = \pi I_{\mathrm{S,scat}}.$$

$$I_{\mathrm{S,scat}} = \frac{F_{\mathrm{scat}}}{\pi} = \frac{pF_{\mathrm{in}}}{\pi} = p\mu_i I_{\mathrm{inc}}(\nu).$$

$$I_{\mathrm{S,scat}}(\theta,\phi,\nu) = p\cos\theta\cos\phi\, I_{\mathrm{inc}}(\nu) \quad (\alpha = 0).$$

Lambert assumption: isotropic outgoing intensity.

The geometric albedo: making the physics explicit

Use (θ, ϕ) only for **position on the planetary surface**. Let $\hat{\mathbf{k}}$ point toward the star and $\hat{\boldsymbol{o}}$ toward the observer. The dimensionless reflection coefficient $p(\theta, \phi; \hat{\mathbf{k}}, \hat{o}, \nu)$ includes directional and multiple scattering; it is not the single-scattering phase function P.

$$I_{\mathrm{S,scat}}(\theta, \phi, \nu) = \mu_i I_{\mathrm{inc}}(\nu)\, p(\theta, \phi; \hat{\mathbf{k}}, \hat{o}, \nu), \qquad \mu_i \equiv \hat{\mathbf{n}} \cdot \hat{\mathbf{k}}.$$

At **full phase** ($\alpha = 0$), the star and observer directions coincide. Thus the incidence and viewing projections are equal:

$$\mu_i = \mu_o \equiv \hat{\mathbf{n}} \cdot \hat{\boldsymbol{o}} = \cos\theta \cos\phi, \qquad p_0(\theta, \phi, \nu) \equiv p(\theta, \phi; \hat{\mathbf{k}}, \hat{o}, \nu).$$

Substituting into the observed-flux integral, $I_{\mathrm{inc}}(\nu)$ cancels:

$$\boxed{A_g(\nu) = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} p_0(\theta, \phi, \nu) \cos^2\phi \cos^3\theta \, \mathrm{d}\theta \, \mathrm{d}\phi}$$

One factor $\cos\phi \cos\theta$ is the incidence projection; the other $\cos\phi \cos^2\theta$ is the viewing projection times the surface-area factor. This is a **surface integral**, not an integral over incident directions (Seager, Eqs. 3.18–3.19, with consistent notation).

A perfect Lambert surface has $p = 1$: energy conservation

Consider one surface element whose normal makes the angle Θ with the direction to the star, and use Seager's description $I_{\mathrm{S,scat}} = p \cos\Theta I_{\mathrm{inc}}(\nu)$.

Definition of p. The dimensionless reflection coefficient is $p(\nu) \equiv I_{\mathrm{S,scat}}(\nu)/[\cos\Theta I_{\mathrm{inc}}(\nu)]$. For a Lambert surface, $p = F_{\mathrm{out}}/F_{\mathrm{in}}$: the fraction of incident energy reflected, distinct from the geometric albedo A_g.

1. **Flux in.** The stellar flux through the element is foreshortened by $\cos\Theta$; with $I_{\mathrm{inc}} \equiv F_{\mathrm{inc}}/\pi$,
$$F_{\mathrm{in}} = \cos\Theta F_{\mathrm{inc}} = \pi \cos\Theta I_{\mathrm{inc}}(\nu).$$
2. **Flux out.** A Lambert surface scatters with the *same* intensity into every outgoing direction, so the flux leaving it is
$$F_{\mathrm{out}} = \int_0^{2\pi} \int_0^{\pi/2} I_{\mathrm{S,scat}} \cos\vartheta \, \sin\vartheta \, \mathrm{d}\vartheta \, \mathrm{d}\varphi = \pi I_{\mathrm{S,scat}} = \pi p \cos\Theta I_{\mathrm{inc}}(\nu).$$
3. **Energy conservation.** A *perfect* Lambert surface absorbs nothing, so $F_{\mathrm{out}} = F_{\mathrm{in}}$:
$$\pi p \cos\Theta I_{\mathrm{inc}} = \pi \cos\Theta I_{\mathrm{inc}} \quad \Longrightarrow \quad \boxed{p = 1}$$

- The two factors of π — the hemispheric integral of $\cos\vartheta$ and the definition $I_{\mathrm{inc}} \equiv F_{\mathrm{inc}}/\pi$ — cancel, which is why p is simply the reflected fraction: a surface that reflects the fraction A has $p = A$.
- This $p = 1$ is what gave $A_g = 2/3$ for the Lambert sphere, and $A_{\mathrm{S}} = A_g\, q = 1$: nothing absorbed.

A_g of a Lambert sphere, and why A_g can exceed 1

For a **Lambert sphere** the intensity is scattered equally in all directions, $p_0(\theta, \phi, \nu) = 1$, and

$$A_g^{\text{Lambert}} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos^2\phi \cos^3\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{1}{\pi} \cdot \frac{\pi}{2} \cdot \frac{4}{3} = \boxed{\frac{2}{3}}$$

- $A_g = 2/3 < 1$ even though a Lambert sphere scatters *all* incident radiation and absorbs none. One third of the light is scattered out of the line of sight: the limb regions are illuminated obliquely *and* seen obliquely, so they contribute less than the corresponding parts of a flat disk.
- By the same definition, A_g *can be greater than 1*: it compares the planet not to a diffusing sphere but to a perfectly diffusing disk. **Enceladus** has $A_g = 1.38$ at visible wavelengths (Verbiscer et al. 2007): fresh ice backscatters strongly, so at $\alpha = 0$ it returns more light toward the observer than a Lambert disk would. Rayleigh-scattering atmospheres also backscatter more than Lambert ($A_g = 3/4$ for a semi-infinite conservative Rayleigh atmosphere).
- Measurements of A_g have been elusive for exoplanets (upper limits from MOST, CoRoT, Kepler; a few detections such as Kepler-7b with $A_g \simeq 0.3$), but A_g remains the standard way to classify theoretical models.

The fraction scattered toward $\alpha = 0$ is A_g/π.

A_g compares the planet with a Lambert disk; it is *not* the fraction of the intercepted energy sent back toward the star. That fraction per steradian (exercise 3.5) follows in three steps.

1. **Power intercepted.** The planet presents its cross section to the star: $P_{\text{inc}} = \pi R_p^2 F_{\text{inc}}$.
2. **Power sent toward the observer.** By definition of A_g, with $\pi I_{\text{inc}} = F_{\text{inc}}$, the flux received at distance $D_\oplus$ at $\alpha = 0$ is $F_\oplus = A_g\,(R_p/D_\oplus)^2 F_{\text{inc}}$. Multiplying by $D_\oplus^2$ turns this into power per unit solid angle leaving the planet in that direction: $(\mathrm{d}P/\mathrm{d}\Omega)_{\alpha=0} = D_\oplus^2 F_\oplus = A_g R_p^2 F_{\text{inc}}$.
3. **The ratio.** Dividing by the intercepted power,

$$\boxed{\frac{1}{P_{\text{inc}}}\left(\frac{\mathrm{d}P}{\mathrm{d}\Omega}\right)_{\alpha=0} = \frac{A_g R_p^2 F_{\text{inc}}}{\pi R_p^2 F_{\text{inc}}} = \frac{A_g}{\pi}\ \text{sr}^{-1}.}$$

- It is a fraction *per steradian*. Energy conservation limits only its integral over all directions ($\leq A_{\text{B}} \leq 1$), which is why a strong backscatterer can have $A_g > 1$.
- Checks: a Lambert disk facing the star ($A_g = 1$) sends $1/\pi$ per sr along its normal, as the cosine law requires; a planet scattering everything isotropically sends $1/(4\pi)$ per sr, i.e. $A_g = 1/4$; a Lambert sphere ($A_g = 2/3$) sends $2/(3\pi)$, which is $8/3$ times the isotropic value — the same $8/3$ as the dayside factor f, from the same double-cos ϑ integral. Enceladus: $1.38/\pi \simeq 0.44$ per sr.

The Bond albedo A_B and the spherical albedo $A_\mathrm{S}(\nu)$

- The **bond albedo** A_B is the fraction of incident stellar *energy* scattered back into space, including radiation into *all* directions and at *all* frequencies. By definition $A_\mathrm{B} \leq 1$.
- The **spherical albedo** $A_\mathrm{S}(\nu)$ is the same quantity **at a specific frequency**. The Bond albedo is the spherical albedo averaged over frequency *weighted by the incident stellar spectrum*:

$$\boxed{A_\mathrm{B} = \frac{\int_0^\infty A_\mathrm{S}(\nu) F_{\mathrm{S},*}(\nu)\,\mathrm{d}\nu}{\int_0^\infty F_{\mathrm{S},*}(\nu)\,\mathrm{d}\nu}}$$

  (Seager writes $A_\mathrm{B} = \int A_\mathrm{S}\,\mathrm{d}\nu$ as shorthand and states that the weighting by the incident radiation is implied; Heng's Eq. 2.10 gives the weighted form explicitly.)
- Consequently A_B depends on the *host star's spectrum*: the same planet has a different Bond albedo around a G star and around an M star, because an M star emits most of its light in the near-IR where the atmosphere may be dark.

To derive $A_\mathrm{S}(\nu)$ we must account for the light scattered at *every* phase angle, not just at $\alpha = 0$. At each α only a portion of the hemisphere facing the observer is illuminated (next slide).

Scattered flux at phase angle α

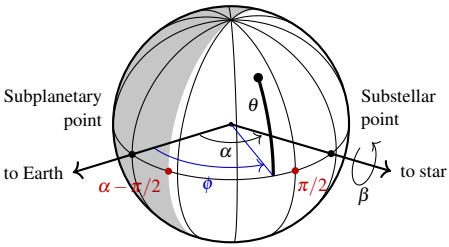


Figure 1: Coordinates for the phase-dependent scattered flux. θ is latitude from the intensity equator, ϕ longitude from the sub-Earth point, α the phase angle, β the azimuth of the observer about the star direction (next slide). After Sobolev (1975); redrawn after Seager (2010), Fig. 3.6.

Measure longitude ϕ from the sub-Earth (subplanetary) point. The substellar point is then at longitude α, and a surface element at (θ,ϕ) is illuminated with the projection $\cos(\alpha-\phi)\cos\theta$ and seen with the projection $\cos\phi\cos\theta$. Only the longitudes $\alpha-\pi/2<\phi<\pi/2$ are both lit and visible. The scattered flux toward the observer is

$$F_{\mathrm{S},\alpha}(\nu)=\int_{\alpha-\pi/2}^{\pi/2}\int_{-\pi/2}^{\pi/2} I_{\mathrm{inc}}(\nu)\,p_\alpha(\theta,\phi,\nu) \times\cos(\alpha-\phi)\cos\phi\cos^3\theta\,\mathrm{d}\theta\,\mathrm{d}\phi,$$

where $p_\alpha\equiv p(\theta,\phi;\hat{\mathbf{k}},\hat{\boldsymbol{o}},\nu)$ is evaluated at phase α. For uniform incident intensity,

$$\boxed{F_{\mathrm{S},\alpha}(\nu)=I_{\mathrm{inc}}(\nu)\,\Psi_\alpha(\nu),}$$

where Ψ_α is the fraction of scattered radiation in the direction of the observer as a function of α. At $\alpha=0$ it reduces to the geometric-albedo integral: $\boxed{\Psi_{\alpha=0}=\pi A_g}$.

The spherical albedo: integrating over all directions

To count the energy scattered into all directions, surround the planet with a sphere and use α as the polar angle and β as the azimuth (azimuthal symmetry assumed). Both are measured at the planet's center: α is the angle between the direction to the star and the direction to a point on the sphere — the phase angle of an observer there — and β the azimuth about the star direction. **Each surface element reflects only into its own outward hemisphere, but the curved lit face as a whole sends light toward every α from 0 to π (with $\Psi_\alpha \to 0$ as $\alpha \to \pi$).** The total scattered energy per unit time per frequency is

$$E_{\rm scat}(\nu) = I_{\rm inc}(\nu) \int_0^{2\pi} \int_0^{\pi} \Psi_\alpha(\nu)\, R_p^2 \sin\alpha \,\mathrm{d}\alpha \,\mathrm{d}\beta = 2\pi R_p^2 I_{\rm inc}(\nu) \int_0^{\pi} \Psi_\alpha(\nu) \sin\alpha \,\mathrm{d}\alpha.$$

The incident energy (Chapter 2) is $E_{\rm inc}(\nu) = \pi R_p^2 \cdot \pi I_{\rm inc}(\nu)$. Their ratio is the spherical albedo:

$$\boxed{A_{\rm S}(\nu) = \frac{E_{\rm scat}(\nu)}{E_{\rm inc}(\nu)} = \frac{2}{\pi} \int_0^{\pi} \Psi_\alpha(\nu) \sin\alpha \,\mathrm{d}\alpha}$$

- $A_{\rm S}$ needs Ψ_α at *all* phase angles. From Earth we cannot see all phases of most planets, so $A_{\rm S}$ (and $A_{\rm B}$) are not directly measurable: we need a way to estimate them from the one thing we can hope to measure near full phase, A_g.
- Note that, in the factor $2/\pi$, the factor of 2 comes from the sphere-to-disk conversion, while the factor of π arises from the flux.

Relating A_S to A_g: the phase integral q

Using $\Psi_{\alpha=0}(\nu) = \pi A_g(\nu)$ (next slide), rewrite the spherical albedo as

$$A_S(\nu) = \frac{\pi A_g(\nu)}{\Psi_{\alpha=0}(\nu)} \cdot \frac{2}{\pi} \int_0^\pi \Psi_\alpha(\nu) \sin\alpha \, \mathrm{d}\alpha \quad \Longrightarrow \quad \boxed{A_S(\nu) = A_g(\nu) q(\nu)}$$

with the **phase integral**

$$q(\nu) \equiv \frac{A_S(\nu)}{A_g(\nu)} = 2 \int_0^\pi \Phi_\alpha(\nu) \sin\alpha \, \mathrm{d}\alpha, \qquad \Phi_\alpha(\nu) \equiv \frac{\Psi_\alpha(\nu)}{\Psi_{\alpha=0}(\nu)}$$

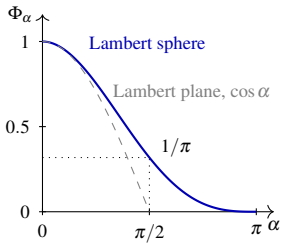
where Φ_α is the **phase function**: the scattered flux versus phase angle normalized to full phase (this is the phase curve).

- A_S can be *estimated* from a measured A_g and a theoretical q. Heng writes the same thing as $A_S = A_g P_g$.
- **Lambert sphere:** $\Phi_\alpha = \frac{1}{\pi}[\sin\alpha + (\pi - \alpha)\cos\alpha]$ (derived on the next slide), so $q = 3/2$ and

  $$A_g = \tfrac{2}{3} A_S, \qquad A_S = A_B = 1, \; A_g = \tfrac{2}{3}.$$

  All light is scattered back to space, so the spherical albedo is 1; the geometric albedo counts only the light scattered toward the observer at $\alpha = 0$ and is less.
- **Rayleigh scattering:** $q = 4/3$; a Lambert *plane* (exercise 3.6): $\Phi_\alpha = \cos\alpha$. Real planets: $q \simeq 1.2$–1.7.

The Lambert sphere: deriving Φ_α and q



Phase function of a Lambert sphere. At quadrature it is only $1/\pi$ as bright as at full phase.

$\Phi_0 = 1$, $\Phi_{\pi/2} = 1/\pi$, $\Phi_\pi = 0$. With $\int_0^\pi \sin^2\alpha\,\mathrm{d}\alpha = \pi/2$ and $\int_0^\pi (\pi-\alpha)\sin\alpha\cos\alpha\,\mathrm{d}\alpha = \pi/4$:

$$q = 2\int_0^\pi \Phi_\alpha \sin\alpha\,\mathrm{d}\alpha = \tfrac{2}{\pi}\left(\tfrac{\pi}{2} + \tfrac{\pi}{4}\right) = \tfrac{3}{2},$$

$$A_S = A_g\, q = \tfrac{2}{3}\cdot\tfrac{3}{2} = 1.$$

In general $\Psi_{\alpha=0} = \pi A_g$. Seager's Eq. (3.30) prints A_g/π, which would make her Eq. (3.31) inconsistent with Eq. (3.29).

For a perfect Lambert surface $p = 1$, and the integral for Ψ_α separates into a latitude and a longitude part:

$$\Psi_\alpha = \int_{-\pi/2}^{\pi/2} \cos^3\theta\,\mathrm{d}\theta \int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha-\phi)\cos\phi\,\mathrm{d}\phi.$$

- **Latitude:** $\int_{-\pi/2}^{\pi/2} \cos^3\theta\,\mathrm{d}\theta = \frac{4}{3}$.
- **Longitude:** with $\cos(\alpha-\phi)\cos\phi = \frac{1}{2}[\cos\alpha + \cos(\alpha-2\phi)]$ over the lit and visible range of width $\pi-\alpha$,

$$\int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha-\phi)\cos\phi\,\mathrm{d}\phi = \frac{1}{2}\left[(\pi-\alpha)\cos\alpha + \left(-\tfrac{1}{2}\sin(\alpha-2\phi)\right)_{\alpha-\pi/2}^{\pi/2}\right]$$

$$= \frac{1}{2}\left[\sin\alpha + (\pi-\alpha)\cos\alpha\right].$$

Hence

$$\Psi_\alpha = \frac{2}{3}\left[\sin\alpha + (\pi-\alpha)\cos\alpha\right], \qquad \Psi_{\alpha=0} = \frac{2\pi}{3} = \pi A_g \;\Rightarrow\; A_g = \frac{2}{3},$$

$$\boxed{\Phi_\alpha = \tfrac{1}{\pi}\left[\sin\alpha + (\pi-\alpha)\cos\alpha\right]}$$

Bond versus geometric albedo for the giant planets

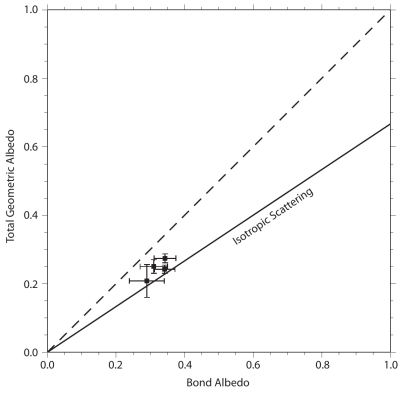


Figure 2: Total (wavelength-integrated) geometric albedo versus Bond albedo for a Lambert sphere (solid, $A_g/A_B = 2/3$) and for Uranus, Neptune, Saturn, Jupiter (points, in order of increasing A_B). The dashed line is $A_g = A_B$. Adapted from Rowe et al. (2006); reproduced from Seager (2010), Fig. 3.7.

- All four giants lie *above* the Lambert (isotropic scattering) line: their atmospheres backscatter more than a Lambert sphere, so $A_g/A_B > 2/3$ ($q < 3/2$).
- The wedge between the Lambert line and $A_g = A_B$ is a useful approximate range for typical optically thick scattering atmospheres. With consistent stellar-spectrum weighting, $A_B > A_g$ corresponds to an effective $q > 1$: these planets generally lie to the right of the dashed line.
- **This is not a universal bound.** Atmospheric depth alone does not guarantee $q > 1$; a sufficiently narrow backscattering peak can give $q < 1$ and $A_g > A_S$ at a given wavelength.
- Practical use: a measured (or upper-limit) A_g of an exoplanet, e.g. the MOST limit $A_g < 0.25$ for HD 209458b, can be converted to a range of A_B, hence to a range of T_{eq}.

The apparent albedo: the one an exoplanet can give us

The geometric, spherical, and Bond albedos are unsatisfactory for exoplanets because *none can be measured*:

- A_g is defined at full phase ($\alpha = 0$), when the planet is behind the star and unobservable (for transiting planets the eclipse depth measured just before/after eclipse is the practical substitute, at a small α).
- A_S and A_B require all phase angles. With direct imaging, phases where the planet projects too close to the star are inaccessible, and the planet does not go through all phases at all unless the orbit is edge-on.

Apparent albedo. The ratio of the scattered flux emerging from the planet toward the observer to the scattered flux from a *perfectly reflecting Lambert sphere* of the same size, at the same phase angle, at the same distance:

$$A_{\rm app}(\alpha,\nu) = \frac{F_{{\rm S},\alpha}(\nu)}{F_{{\rm S},\alpha}^{\rm Lambert}(\nu)} = \frac{A_g(\nu)\,\Phi_\alpha(\nu)}{\frac{2}{3}\,\Phi_\alpha^{\rm Lambert}} \qquad \text{(exercise 3.9)}.$$

- It is the planet's reflectance in the direction of the observer at a given time, a function of phase angle and frequency.
- It is the only albedo that can be derived from a *single* observation of a planet of known size — a directly imaged planet at known separation and phase.

The emergent flux

The emergent surface flux follows from the flux definition (Chapter 2) applied to the outgoing hemisphere, $F = 2\pi \int_0^1 \mu\, I(0,\mu)\, \mathrm{d}\mu$ (no incoming radiation at the top in the thermal infrared):

$$\boxed{F(0,\nu) = 2\pi \int_0^1 \int_0^\infty S(\tau_\nu, \mu, \nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu}$$

- If the source function $S(\tau,\mu,\nu)$ is *known*, the emergent intensity and the emergent flux — the planet's spectrum — can be computed directly from these two integrals. The μ integral of $\mathrm{e}^{-\tau/\mu}$ produces the exponential integral $E_2(\tau)$, so $F(0,\nu) = 2\pi \int_0^\infty S(\tau_\nu) E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$ for an isotropic S.
- For $S = B(T(\tau))$ (pure absorption, LTE) this is the standard *forward model* of an emission spectrum: choose $T(\tau_\nu)$, i.e. a temperature–pressure profile and an opacity function, and integrate.
- In many cases, however, this straightforward solution is not possible. The main complication is that the source function itself depends on $I(\tau,\mu,\nu)$, the quantity we are trying to solve for.

Planet-Star Flux Ratios

An estimate of the planet-to-star flux ratio tells us whether a given planet and star can be detected with a given telescope, at which wavelengths, and with what precision. Previous slides gave us the ingredients: surface fluxes, the flux at Earth, temperatures, and albedos. We now assemble them.

- Assume star and planet radiate as blackbodies (a first estimate; brightness temperatures refine it).
- A planet has **two spectral peaks**: a short-wavelength peak of *scattered starlight*, at the same wavelength as the star, and a long-wavelength peak of *thermal emission* at the planet's own temperature.
- During transit a third quantity appears: the *transmission* flux ratio, the fraction of starlight blocked by the atmospheric annulus.

The planet flux at Earth is $F_{\oplus,p} = F_{\mathrm{S},p}(R_p/D_\oplus)^2$ and the stellar flux at Earth is $F_{\oplus,*} = F_{\mathrm{S},*}(R_*/D_\oplus)^2$, so the distance cancels:

$$\boxed{\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{\mathrm{S},p}R_p^2}{F_{\mathrm{S},*}R_*^2}}$$

valid for total fluxes and for frequency-dependent fluxes alike.

Spectra of the Solar System seen from 33 light years

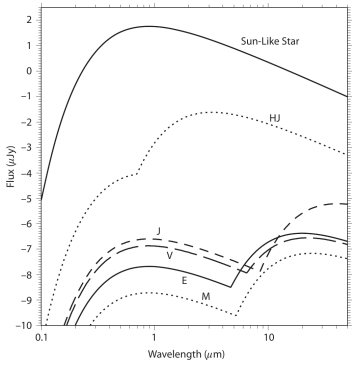


Figure 3: Approximate blackbody spectra (in 10^{-26} W m^{-2} Hz^{-1}) of the Sun (5750 K), Jupiter, Venus, Earth, Mars, and a representative hot Jupiter, seen from 33 light years (10 pc). Reproduced from Seager (2010), Fig. 3.8.

- Each planet shows the two peaks: reflected light near 0.5 μm (shape of the stellar spectrum, scaled by $A_g R_p^2/a^2$) and thermal emission peaking at $\lambda_{\max} \simeq 2898\,\mu\mathrm{m\,K}/T$: $\sim 10\,\mu$m for Earth and Venus, $\sim 25\,\mu$m for Jupiter, $\sim 2\,\mu$m for a representative hot Jupiter.
- The planet-star contrast is large at *all* wavelengths, but the thermal peak of a hot Jupiter reaches within 10^{-3} of the star, the Earth's thermal peak within 10^{-7}, while in the visible the contrast is 10^{-9}–10^{-10}.
- Venus is brighter than Earth in reflected light (high albedo, closer to the Sun) but similar in the thermal infrared: its cloud tops radiate at $T_{\mathrm{eff}} \simeq 230$ K.
- This high contrast is *the* primary challenge for detecting radiation from exoplanets.

Thermal emission flux ratio

On the illuminated side, equate the planet's thermal emission with the absorbed stellar radiation — the energy balance with the redistribution factor f, $4\pi R_p^2 F_{\mathrm{S},p}/f = (1-A_\mathrm{B})F_{\mathrm{S},*}(R_*/a)^2\pi R_p^2$ — to get the *total* flux ratio

$$\boxed{\frac{F_{\oplus,p}}{F_{\oplus,*}} = \frac{F_{\mathrm{S},p}R_p^2}{F_{\mathrm{S},*}R_*^2} = \left(\frac{R_p}{a}\right)^2 \frac{f}{4}(1-A_\mathrm{B})}$$

- Here f is the correction factor to 4π that describes the unknown redistribution of the absorbed stellar radiation, and the fluxes are total fluxes, integrated over all frequencies.
- **The ratio depends on R_p/a, not on R_p/R_***: the planet reradiates the fraction $\pi R_p^2/4\pi a^2$ of the stellar luminosity it intercepts, times $(1-A_\mathrm{B})$, times the redistribution factor.
- For a hot Jupiter at $a = 0.05$ AU with $A_\mathrm{B} = 0.1$ and $f = 1$: $(R_\mathrm{J}/a)^2 \times 0.9/4 \simeq 2\times 10^{-5}$ bolometrically. Yet the observed ratio in an infrared band is $\sim 10^{-3}$, because the planet's emission is concentrated at long wavelengths where the star is faint — which is why we need a frequency-dependent estimate.

Thermal emission flux ratio at a given frequency

A more useful planet-star flux ratio estimate is **frequency dependent**, because telescope observations are limited to a fixed frequency or wavelength range. Approximate the star and planet surface fluxes by blackbody fluxes $\pi B(T,\nu)$, with T the effective temperature of the star and the equilibrium temperature of the planet:

$$\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{\mathrm{S},p}(\nu)R_p^2}{F_{\mathrm{S},*}(\nu)R_*^2} = \frac{\mathrm{e}^{h\nu/k_\mathrm{B}T_{\mathrm{eff},*}}-1}{\mathrm{e}^{h\nu/k_\mathrm{B}T_{\mathrm{eq},p}}-1}\,\frac{R_p^2}{R_*^2}$$

- The ν^3 prefactors of the two Planck functions cancel; only the exponential factors and the area ratio remain.
- In the Wien tail of the planet ($h\nu \gg k_\mathrm{B}T_\mathrm{eq}$) the ratio is $\simeq \mathrm{e}^{-h\nu(1/k_\mathrm{B}T_\mathrm{eq}-1/k_\mathrm{B}T_{\mathrm{eff},*})}(R_p/R_*)^2$, exponentially small; it rises steeply toward the infrared and saturates in the Rayleigh–Jeans regime (next slide).
- Note that **it is now** R_p/R_* **that matters, not** R_p/a: at a fixed planet temperature, a smaller star gives a larger contrast. This is the same "M-dwarf opportunity" as for transits.
- Example: HD 189733b at 8 μm with $R_p/R_* = 0.155$, $T_\mathrm{eq} = 1200\,\mathrm{K}$, $T_{\mathrm{eff},*} = 5000\,\mathrm{K}$ gives $(\mathrm{e}^{3.6}-1)/(\mathrm{e}^{1.5}-1)\times 0.024 = 0.0034$, close to the observed eclipse depth of 0.34% (Lecture 1).

The Rayleigh–Jeans limit

At mid-infrared wavelengths a further simplification applies: for reasonable planet and star temperatures both blackbody fluxes are in the **Rayleigh–Jeans** region, $h\nu/k_{\mathrm{B}}T \ll 1$. With $\mathrm{e}^{x} - 1 \simeq x$ the Planck function becomes

$$\pi B(T,\nu) \simeq \pi \,\frac{2\nu^{2}}{c^{2}}\, k_{\mathrm{B}} T,$$

linear in T, and the flux ratio becomes

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{\mathrm{S},p}(\nu) R_{p}^{2}}{F_{\mathrm{S},*}(\nu) R_{*}^{2}} \simeq \frac{T_{\mathrm{eq},p}}{T_{\mathrm{eff},*}} \left(\frac{R_{p}}{R_{*}}\right)^{2}}$$

in the frequency range where the approximation is valid.

- This is the *maximum* contrast the thermal emission can reach: the temperature ratio times the area ratio.
- **Hot Jupiter:** $(1500/5800) \times (0.1)^{2} \simeq 2.6 \times 10^{-3}$, approached already at $\lambda \gtrsim 10\,\mu\mathrm{m}$ (for $T_{p} = 1500\,\mathrm{K}$, $h\nu/k_{\mathrm{B}}T = 1$ at $\lambda = 9.6\,\mu\mathrm{m}$).
- **Earth:** $(255/5800) \times (0.0092)^{2} \simeq 3.7 \times 10^{-6}$, reached only at $\lambda \gtrsim 100\,\mu\mathrm{m}$; at $10\,\mu\mathrm{m}$ the actual ratio is $\sim 10^{-7}$ because Earth is still on the Wien side.
- The Rayleigh–Jeans form makes explicit why the infrared is the place to detect thermal emission: the contrast saturates at T_{p}/T_{*} (independent of wavelength) instead of decaying exponentially.

Using brightness temperatures instead of blackbodies

Real planets and stars deviate from blackbodies, sometimes strongly (the M-dwarf spectrum of last lecture, molecular bands in the planet). In the Rayleigh–Jeans regime, the flux ratio can be expressed in terms of the **brightness temperatures** at each frequency:

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \frac{F_{\mathrm{S},p}(\nu) R_p^2}{F_{\mathrm{S},*}(\nu) R_*^2} = \frac{T_{\mathrm{b},p}(\nu)}{T_{\mathrm{b},*}(\nu)} \left(\frac{R_p}{R_*}\right)^2} \qquad \text{(Rayleigh–Jeans regime)},$$

and, outside the Rayleigh–Jeans regime, the ratio of Planck functions evaluated at $T_{\mathrm{b},p}(\nu)$ and $T_{\mathrm{b},*}(\nu)$.

- This is exactly how a measured **secondary-eclipse depth is turned into a dayside brightness temperature**: with R_p/R_* from the transit and a stellar model for $T_{\mathrm{b},*}(\nu)$, invert the Planck function for $T_{\mathrm{b},p}(\nu)$.
- The stellar brightness temperature matters: at 3.6–8 μm a K or M dwarf has $T_{\mathrm{b},*}$ well below its T_{eff} because of molecular absorption, which raises the apparent planet-to-star contrast.
- $T_{\mathrm{b},p}(\nu)$ across several bands maps out the emission spectrum: bands with lower T_{b} are absorption features formed high in a (normally stratified) atmosphere; bands with higher T_{b} indicate a temperature inversion.

Scattered-light flux ratio at full phase ($\alpha = 0$)

Scattered starlight is typically at visible wavelengths (extending to longer wavelengths for colder planets). At $\alpha = 0$, start from the thermal flux ratio and replace the absorbed-and-reradiated factor by a scattered one:

$$\frac{f}{4}(1 - A_B) \quad \longrightarrow \quad \frac{g}{4} A_S(\nu),$$

where $A_S(\nu)$ is the spherical albedo (we want a wavelength-dependent ratio; monochromatic scattering assumed). Here, like f, g **is a correction factore to 4π: the solid angle into which the incident radiation is scattered compared to 4π**. Scattered light cannot be advected around the planet; it leaves the illuminated hemisphere, with a directional pattern. Thus

$$\frac{F_{\oplus,p}}{F_{\oplus,*}} = \left(\frac{R_p}{a}\right)^2 \frac{f}{4}(1 - A_B) \quad \Rightarrow \quad \frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = \left(\frac{R_p}{a}\right)^2 \frac{g}{4} A_S(\nu).$$

The critical step is g: the ratio of energy emitted at zero phase angle to that emitted at all phase angles. With $E_{\rm scat}$ from the spherical-albedo derivation and Ψ_α the fraction scattered toward the observer,

$$\frac{g}{4\pi} = \frac{E_{{\rm scat},\alpha=0}}{E_{\rm scat}} = \frac{\Psi_{\alpha=0}(\nu)}{2\pi \int_0^\pi \Psi_\alpha(\nu) \sin\alpha \, {\rm d}\alpha} = \frac{1}{\pi q(\nu)} = \frac{A_g(\nu)}{\pi A_S(\nu)} \quad \Longrightarrow \quad \boxed{g = \frac{4A_g(\nu)}{A_S(\nu)}}$$

using $\Psi_{\alpha=0} = \pi A_g$, the phase function Φ_α, and the phase integral $q = A_S/A_g$.

Scattered-light flux ratio: the result and numbers

Substituting g, the spherical albedo cancels and the full-phase visible flux ratio is

$$\boxed{\frac{F_{\oplus,p}(\nu)}{F_{\oplus,*}(\nu)} = A_g(\nu)\,\frac{R_p^2}{a^2}}$$

- Note that the visible flux ratio at full phase involves the *geometric* albedo, which describes the light scattered toward the observer at $\alpha = 0$, not the spherical or Bond albedo, which describe the energy scattered in all directions. That is precisely what g converts.
- The ratio falls as a^{-2} and does not involve R_* at all (for a fixed stellar flux at the planet, a bigger star only means a bigger a for the same temperature).
- **Jupiter:** $A_g(500\,\mathrm{nm}) = 0.5$, $a = 5.2$ AU, $R_p = R_\mathrm{J}$ gives $F_{\oplus,p}/F_{\oplus,*} = 4.4 \times 10^{-9}$.
- **Scaled to 0.05 AU:** 10^4 times higher, 4.4×10^{-4}. However, the actual hot Jupiters are darker: Jupiter is bright because of its icy NH_3 and H_2O clouds, whereas hot Jupiters lack bright clouds and have strong Na/K absorption, so $A_g \lesssim 0.1$ and the ratio is $\lesssim 10^{-4}$ — consistent with the Kepler and MOST upper limits and detections.
- **Earth:** $A_g(500\text{–}800\,\mathrm{nm}) = 0.2$, $a = 1$ AU: 2.1×10^{-10} at $\alpha = 0$ (the number quoted in Lecture 1).

Transmission flux ratio

During transit, starlight passes through the planet atmosphere and picks up its spectral features: the **transmission spectrum**. What is measured is the total (star + planet) flux in transit divided by the out-of-transit flux. Let R_p be the radius where the planet is opaque at all wavelengths of interest and A_H the additional radial extent of the atmosphere. Splitting the transmission spectrum ($F_{\text{in trans},\oplus}$) into the unobscured stellar disk (I_*), the opaque planet ($I_p = 0$), and the atmospheric annulus (I_{atm}):

$$F_{\text{in trans},\oplus}(\nu) = \int_0^{2\pi}\int_{(R_p+A_H)/D_\oplus}^{R_*/D_\oplus} I_* \cos\omega \sin\omega \,\mathrm{d}\omega\,\mathrm{d}\phi + \int_0^{2\pi}\int_{R_p/D_\oplus}^{(R_p+A_H)/D_\oplus} I_{\text{atm}} \cos\omega \sin\omega \,\mathrm{d}\omega\,\mathrm{d}\phi.$$

Dividing by the out-of-transit stellar flux $F_{*,\oplus} = F_{\text{S},*}(R_*/D_\oplus)^2$ and assuming $A_H \ll R_p$,

$$\boxed{\frac{F_{\text{in trans},\oplus}(\nu)}{F_{*,\oplus}(\nu)} = 1 - \left(\frac{R_p}{R_*}\right)^2 - \frac{2R_pA_H}{R_*^2}\left[1 - \frac{F_{\text{S},p,\text{trans}}(\nu)}{F_{\text{S},*,\text{trans}}(\nu)}\right]}$$

- (1) First term: the star, (2) second: the opaque planetary disk (the **transit depth**), (3) third: the annulus $2\pi R_p A_H$ over the stellar disk πR_*^2, times the fraction of starlight the annulus *blocks*. $F_{\text{S},p,\text{trans}}/F_{\text{S},*,\text{trans}} = 1 - e^{-\tau}$ is the transmitted fraction, computed from the slab geometry.

How large is the transmission signal?

The maximum signal occurs where the annulus is completely opaque ($F_{\mathrm{S},p,\mathrm{trans}} = 0$):

$$\frac{F_{p,\oplus,\mathrm{trans}}(\nu)}{F_{*,\oplus}(\nu)} = \frac{2R_pA_H}{R_*^2}\left[1 - \frac{F_{\mathrm{S},p,\mathrm{trans}}}{F_{\mathrm{S},*,\mathrm{trans}}}\right] \approx \frac{2R_pA_H}{R_*^2}$$

- The choice of A_H does not matter as long as it is large enough: above some height there is no atmosphere left to attenuate, the transmitted fraction is 1, and the bracket vanishes. What matters is the height at which the slant path becomes opaque, a few scale heights above the opaque radius.
- **Hot Jupiter** around a Sun-sized star: an H/He atmosphere with $T_{\mathrm{eq}} \simeq 1000\,\mathrm{K}$ has a scale height $H \simeq 500\,\mathrm{km}$ (Chapter 9). Taking the atmosphere to extend $A_H = 5H$: $2R_pA_H/R_*^2 = 2 \times (7.1 \times 10^4) \times (2.5 \times 10^3)/(7 \times 10^5)^2 \simeq 7 \times 10^{-4}$, i.e. of order 10^{-3} (Seager quotes 1.5×10^{-3}).
- One part per thousand may seem very small. In fact it is a *large* number in the field of exoplanet atmosphere detection: compare with the $\lesssim 10^{-4}$ reflected-light and $\sim 10^{-3}$ thermal ratios above, and note that it requires no separation of the planet's light from the star's, only precise photometry in and out of transit.

Transmission signals of cooler planets, and the three channels

- **Jupiter analog** (a Jupiter-sized planet in a Jupiter-like orbit, $T_{\rm eq} \simeq 120\,{\rm K}$, $H = 24\,{\rm km}$): the estimated transmission strength is only ~ 4–7×10^{-5}. Cold planets have small scale heights and thus weak transmission spectra, even though their transit depth is the same.
- **Earth twin** ($H = 8\,{\rm km}$, N_2 atmosphere): $\sim 2 \times 10^{-7}$ (Lecture 1). This is why terrestrial-planet transmission spectroscopy targets M dwarfs, where R_*^2 is 25–100 times smaller.
- The transmission technique has been very successful for hot Jupiters around bright stars: Na, K, H_2O, CH_4, CO, CO_2, He, and escaping H have all been identified this way, and JWST now routinely delivers transmission spectra of sub-Neptunes.

The three observational channels and what controls each:

channel	flux ratio	key planet property	geometric factor
thermal emission	$(T_{{\rm b},p}/T_{{\rm b},*})(R_p/R_*)^2$	temperature $T_{\rm b}(\nu)$	$(R_p/R_*)^2$
reflected light	$A_g R_p^2/a^2$	geometric albedo $A_g(\nu)$	$(R_p/a)^2$
transmission	$2R_p A_H/R_*^2$	scale height H	R_p/R_*^2

Each channel favors different planets: emission and transmission favor hot, large planets around small stars; reflection favors close-in planets but is limited by their low albedos.

Visible-wavelength phase curves

An exoplanet goes through illumination phases like the Moon. We want the flux ratio not only at $\alpha = 0$ but as a function of α. Recall the normalized phase function

$$\Phi_\alpha(\nu) = \frac{\Psi_\alpha(\nu)}{\Psi_{\alpha=0}(\nu)}, \qquad \Psi_\alpha(\nu) = \int_{\alpha-\pi/2}^{\pi/2}\int_{-\pi/2}^{\pi/2} p_\alpha(\theta,\phi,\nu)\cos(\alpha-\phi)\cos\phi\cos^3\theta\,\mathrm{d}\theta\,\mathrm{d}\phi.$$

The Lambert sphere is the one case with an analytic phase curve. With $p = 1$ and uniform incident intensity, the θ integral gives $4/3$ and the ϕ integral gives $\int_{\alpha-\pi/2}^{\pi/2}\cos(\alpha-\phi)\cos\phi\,\mathrm{d}\phi = \frac{1}{2}[\sin\alpha + (\pi-\alpha)\cos\alpha]$, so after normalizing to $\alpha = 0$ (where the integral is $\pi/2$),

$$\boxed{\Phi_\alpha = \frac{1}{\pi}\left[\sin\alpha + (\pi-\alpha)\cos\alpha\right]}$$

frequency independent, with $\Phi_0 = 1$, $\Phi_{\pi/2} = 1/\pi$, $\Phi_\pi = 0$.

- At quadrature ($\alpha = 90°$) a Lambert sphere is $1/\pi \simeq 0.32$ as bright as at full phase, *less than half* (exercise 3.7): the visible illuminated half-disk is lit obliquely near the terminator and its brightest region (the substellar point) sits at the limb, seen at grazing angle.
- The phase integral follows: $q = 2\int_0^\pi \Phi_\alpha \sin\alpha\,\mathrm{d}\alpha = 3/2$.

Phase curves of Solar System bodies

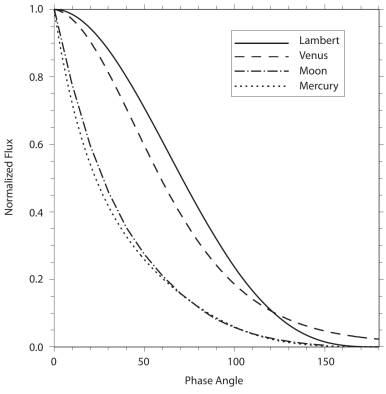


Figure 4: Visible-wavelength phase curves of Venus, Mercury, and the Moon, with the Lambert sphere (rightmost solid line). From formulae in Cox (2000); reproduced from Seager (2010), Fig. 3.9.

- **Venus**, with its thick cloud deck, follows the Lambert curve closely out to $\alpha \sim 100^\circ$ and then falls faster (forward scattering by cloud droplets brightens it again at very large α, not shown).
- The **Moon** and **Mercury** — airless, rough, regolith-covered — fall off much faster: their surfaces backscatter (the *opposition surge* near $\alpha = 0$) and shadowing in the rough surface darkens the oblique views. At quadrature the Moon is only ~ 0.1 of its full-phase brightness, not 0.32.
- Lesson: the phase curve shape encodes the *scattering physics* (clouds versus bare surface, particle size, surface roughness). For exoplanets it is a second observable beyond A_g.

Flux ratio at all phases and an Earth example

The normalized phase curve combined with the full-phase flux ratio gives the planet-star flux ratio at any phase:

$$\frac{F_{p,\oplus,\alpha}(\nu)}{F_{*,\oplus}(\nu)} = A_g(\nu)\,\frac{R_p^2}{a^2}\,\Phi_\alpha(\nu)$$

Earth orbiting the Sun as an exoplanet.

- At $\alpha = 0$ with $A_g(500\text{–}800\,\text{nm}) = 0.2$, $a = 1$ AU, $R_p = R_\oplus$: 2.1×10^{-10}. Earth is almost ten billion times fainter than the Sun — and $\alpha = 0$ is when the planet is directly behind the star.
- At **maximum elongation** ($\alpha = 90°$), the natural configuration for direct imaging, approximating Earth as a Lambert sphere: $\Phi_{90°} = 1/\pi$ and the ratio is 1.2×10^{-10}, a factor π lower. (Earth's real phase curve is not Lambertian: ocean glint and forward-scattering clouds modify it.)
- For an edge-on orbit the full range $0 < \alpha < 180°$ is swept once per orbit; the amplitude of the reflected-light phase curve is $A_g R_p^2/a^2$ and its *shape* constrains Φ_α, hence q and A_S. For inclination i, the accessible range is $90° - i \leq \alpha \leq 90° + i$ (exercise 3.8).

Infrared phase curves

Phase variation also occurs at *infrared* wavelengths, from thermal emission, and not just at visible wavelengths from scattered starlight. Thermal phase variation arises when the emission is dominated by reemitted absorbed starlight rather than by internal energy.

- The prime examples are the **tidally locked** (synchronously rotating) hot Jupiters, hot Neptunes, and hot Earths with $a < 0.05$ AU, strongly heated by their stars. They present the same face to the star at all times, just as the Moon does to Earth.
- The permanent day and night sides set up a temperature gradient that atmospheric circulation may or may not even out. How the absorbed stellar energy is circulated is a complex topic (Chapter 10).

Simplest case: a hot tidally locked planet where the incident stellar energy is absorbed in one layer, instantaneously heats that surface element, and is reemitted, with no advection. Thermal emission is isotropic, so each element reradiates like a Lambert scatterer and the problem is solved in analogy with the Lambert sphere: the normalized thermal phase curve is again

$$\Phi_\alpha = \tfrac{1}{\pi}\left[\sin\alpha + (\pi - \alpha)\cos\alpha\right],$$

peaking at secondary eclipse and vanishing at transit (no nightside emission at all), with the dayside temperature corresponding to $f' = 2/3$.

Infrared phase curves: the general case

Beyond the simplest case, if the thermal flux distribution over the planet's surface is available from a model atmosphere or a circulation model, the thermal phase curve is still computed from the planet-star flux ratio, with the phase-dependent planetary flux

$$F_{\alpha,\mathrm{thermal}}(\nu) = \int_{-\pi/2}^{\pi/2}\int_{-\pi/2}^{\pi/2} I_{\alpha,\mathrm{thermal}}(\theta,\phi,\nu)\cos\phi\cos^2\theta\,\mathrm{d}\theta\,\mathrm{d}\phi,$$

integrated over the hemisphere facing the observer at phase α (the subscript "thermal" denotes the thermal-infrared component of the intensity), and the flux ratio follows with R_p^2/R_*^2 and the stellar flux at that frequency.

- The shape of $I_{\alpha,\mathrm{thermal}}(\theta,\phi)$ encodes the longitudinal temperature map: a hot spot shifted east of the substellar point produces a phase-curve peak *before* secondary eclipse (Lecture 1).
- The nightside flux sets the minimum of the phase curve near transit; the day–night amplitude measures how efficiently heat is redistributed, i.e. the effective f'.
- Real hot Jupiters (HD 189733b, HD 209458b) show amplitudes reduced relative to the no-redistribution Lambert curve and offset peaks: heat is carried to the nightside by winds. Ultra-hot Jupiters and planets on eccentric orbits (HD 80606b) show the full range from strong to weak redistribution.

[Radiative transfer] - the emergent spectrum

We have built the tools: optical depth, opacity, LTE, the source function, and the transfer equation with its formal solution. We have then used them for the albedos, the flux ratios and the phase curves. We now turn to the emergent spectrum itself:

1. **What does a spectrum tell us?** Why lines appear in absorption or in emission, at which depth they form, and what they reveal about the vertical temperature structure (Seager Ch. 6).
2. **How do we solve the equation** for transmitted, emitted and scattered light? Lecture 6 continues with the practical checklist and the two-stream approximation (Heng Ch. 3).

Working assumptions throughout: a one-dimensional, time-independent, plane-parallel atmosphere made of homogeneous layers, in hydrostatic equilibrium, with a *known* temperature–pressure profile. **How the temperature–pressure profile is itself determined is the subject of a later lecture (Heng Ch. 4, Seager Ch. 9).**

The equation we solve, and its sign conventions

Along a ray, the intensity loses κI and gains the emission coefficient ε. With z the height and $\mu = \cos\theta$ measured from the upward normal ($\mathrm{d}z/\mathrm{d}s = \mu$), the RT equation becomes

$$\mu \frac{\mathrm{d}I(z,\mu,\nu)}{\mathrm{d}z} = -\kappa(z,\nu)\, I(z,\mu,\nu) + \varepsilon(z,\mu,\nu).$$

Measuring the optical depth *downward* from the top, $\mathrm{d}\tau_\nu = -\kappa\,\mathrm{d}z$, and writing $S = \varepsilon/\kappa$ turns this into the form used by both books:

$$\boxed{\mu \frac{\mathrm{d}I}{\mathrm{d}\tau_\nu} = I - S}$$

- $\mu > 0$: outgoing (upward) rays; $\mu < 0$: incoming (downward) rays. $\tau_\nu = 0$ at the top of the atmosphere.
- All quantities are monochromatic; the subscript ν is dropped when no confusion arises.
- Everything in this lecture is about choosing S and solving this one line.

A planetary spectrum has two peaks

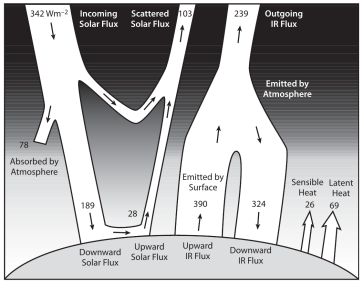


Figure 5: Energy balance of Earth's atmosphere (global means, in $\mathrm{W\,m^{-2}}$). Seager (2010), Fig. 6.1.

Latent heat: evaporation absorbs energy and cools the surface. Water vapor carries this energy upward; condensation releases it, warming the air. The energy is stored through a *phase change*, rather than a temperature rise.

- A **visible peak** of scattered starlight and an **infrared peak** of thermal emission.
- Thermal emission has two sources: absorbed stellar energy that is reradiated, and *interior* energy (residual heat of formation, radioactive decay). For hot Jupiters the first overwhelms the second.
- Reading the figure: of $342\,\mathrm{W\,m^{-2}}$ incident, 103 are scattered back ($A_\mathrm{B} = 0.30$) and 239 leave as infrared, i.e. $\sigma T^4 = 239$ gives $T = 255\,\mathrm{K} = T_\mathrm{eq}$.
- The surface emits $390\,\mathrm{W\,m^{-2}}$ ($= \sigma\,288^4$) because the atmosphere returns $324\,\mathrm{W\,m^{-2}}$ downward: the greenhouse effect, written in fluxes.

Sensible heat: the warm surface heats the air; turbulence and convection carry this heat upward.

Continuum, absorption lines, emission lines

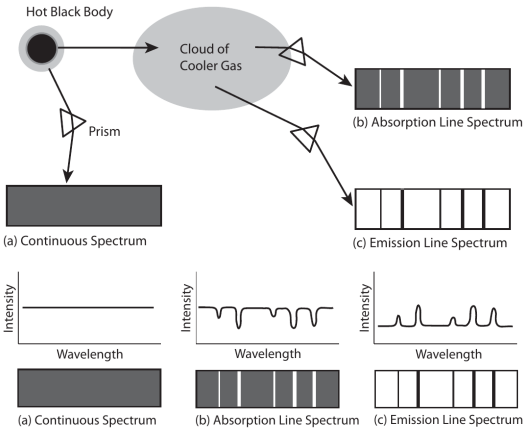


Figure 6: Origin of the three kinds of spectra (top) and what they look like over a narrow wavelength range (bottom). Seager (2010), Figs. 6.2 and 6.3.

- **Continuous spectrum:** a black body; the spectrum depends only on its temperature.
- **Absorption lines:** a cooler gas in front of the hot black body takes "bites" out of the continuum at the wavelengths its atoms and molecules absorb. The absorbed energy is re-emitted isotropically, so little of it returns to the line of sight.
- **Emission lines:** the same gas seen *without* the hot background. Only the re-emitted photons reach us, and they appear at exactly those wavelengths.
- What matters is not the gas alone but the gas *compared with what lies behind it*.

A layered atmosphere: the temperature gradient decides

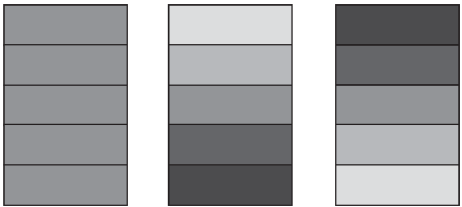


Figure 7: A five-layer atmosphere; darker means hotter. Left: isothermal. Middle: hot at depth, cooler above. Right: a temperature inversion. Seager (2010), Fig. 6.4.

Treat each layer as a homogeneous slab with its own temperature and the deepest layer as a black body.

- **Temperature decreasing outward:** every layer is a cool cloud in front of a hotter one $\Rightarrow$ an **absorption spectrum**. Each layer adds its own bites.
- **Temperature inversion** (hotter upward) $\Rightarrow$ an **emission spectrum**.
- **Isothermal:** no cooler gas in front, no hotter gas behind $\Rightarrow$ a black-body continuum with *no* features at all, however large the opacity.

The last point is counter-intuitive and worth proving (next section): spectral features measure temperature *contrasts*, not the mere presence of absorbers.

Transmission and scattered-light spectra

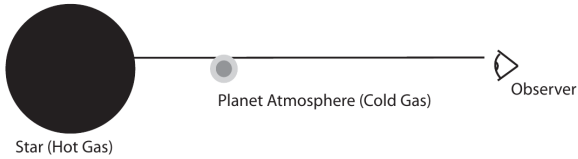


Figure 8: Origin of a transit transmission spectrum: hot stellar light through the cool planetary atmosphere. Seager (2010), Fig. 6.5.

- **Transmission:** the star is the hot black body, the planet's limb is the cool cloud ⇒ always an absorption spectrum. It is isolated by dividing the in-transit spectrum by the out-of-transit stellar spectrum.
- **Scattered light** (visible, near-IR): Seager prefers "scattered" to "reflected", because the light is refracted, diffracted and Rayleigh-scattered rather than bounced off a surface. To first order Earth's visible spectrum is sunlight that crossed the atmosphere *twice* (down to the ground or a cloud deck and back up), a weighted mix of clear and cloudy paths.

Isothermal Atmosphere: layer by layer

Take an n-layer LTE atmosphere whose deepest layer n is optically thick, so $I_n = B_n$. With Kirchhoff's law $\varepsilon_n = \kappa_n B_n$,

$$\mu \frac{\mathrm{d}I_n}{\mathrm{d}z} = \varepsilon_n - \kappa_n I_n = \kappa_n (B_n - B_n) = 0.$$

The intensity entering layer $n-1$ is therefore B_n, and in that layer

$$\mu \frac{\mathrm{d}I_{n-1}}{\mathrm{d}z} = \kappa_{n-1} (B_{n-1} - B_n) = 0 \qquad \text{because } T_{n-1} = T_n.$$

Repeating up to the top, $I_0 = \cdots = I_{n-1} = I_n = B$.

- Absorption and re-emission still happen in every layer, but they occur at the *same temperature* and hence at the same wavelengths: every photon removed is replaced by one of the same frequency.
- Note what the proof used: only the equality of temperatures, not the size of κ_ν. A strongly absorbing and a weakly absorbing wavelength give the same $I = B$.

Isothermal Atmosphere: from the formal solution

For an emitting atmosphere under **LTE conditions**, without scattering and with no radiation entering from above, the emergent intensity is (Lecture 3)

$$I(0,\mu,\nu) = \int_0^\infty B\big(T(\tau_\nu),\nu\big)\,\mathrm{e}^{-\tau_\nu/\mu}\,\frac{\mathrm{d}\tau_\nu}{\mu}.$$

If T is constant, B leaves the integral, and $\int_0^\infty \mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu = 1$, so

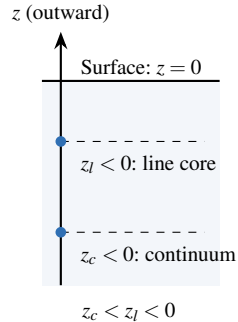
$$I(0,\mu,\nu) = B(T,\nu), \qquad F(0,\nu) = \pi B(T,\nu).$$

- The kernel $\mathrm{e}^{-\tau/\mu}/\mu$ is a *normalized weighting function*: the emergent intensity is a weighted average of B over depth, with weights concentrated near $\tau \sim \mu$.
- A weighted average of a constant is that constant. Only when B varies with depth can different wavelengths, weighted over different depths, come out different.

Non-isothermal Atmosphere: a temperature gradient creates lines

Compare a wavelength in the line core, ν_l, with one in the nearby continuum, ν_c. By definition $\kappa(\nu_l) > \kappa(\nu_c)$.

- From $\mathrm{d}\tau_\nu = -\kappa_\nu \,\mathrm{d}z$ (z increases outward), the same value $\tau_\nu \approx 1$ is reached *higher up* in the line core than in the continuum: $z_c < z_l < 0$ ($|z_l| < |z_c|$).
- The emergent intensity is essentially B at $\tau_\nu \approx \mu$ (Eddington–Barbier). So the line core samples higher layers, the continuum deeper ones.
- **Temperature decreasing outward:** $T(z_l) < T(z_c)$, so $I(\nu_l) < I(\nu_c)$ — an **absorption line**.
- **Temperature increasing outward (inversion):** $T(z_l) > T(z_c)$, so $I(\nu_l) > I(\nu_c)$ — an **emission line**.



A spectrum across a line is therefore a *scan in altitude*: the wings, with lower opacity, look deeper, and the core looks highest.[1]

[1] Seager's text at this point reads "$I(\tau_\nu, \mu, \nu_l) < I(\tau_\nu, \mu, \nu_l)$"; the right-hand side should be the continuum, ν_c.

Non-isothermal Atmosphere: the emergent flux as a depth-weighted B_ν

Integrating the emergent intensity over the outgoing hemisphere,

$$F(0,\nu) = 2\pi \int_0^1 \int_0^\infty B(\tau_\nu,\nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu = 2\pi \int_0^\infty B(\tau_\nu,\nu)\, E_2(\tau_\nu)\, \mathrm{d}\tau_\nu,$$

using $\int_0^1 \mathrm{e}^{-\tau/\mu}\, \mathrm{d}\mu = E_2(\tau)$, the exponential integral $E_n(x) = \int_1^\infty t^{-n} \mathrm{e}^{-xt}\, \mathrm{d}t$ (Seager Eq. 6.26).

- $2E_2(\tau)$ is the **flux weighting function**: it is normalized ($\int_0^\infty 2E_2\, \mathrm{d}\tau = 1$), equals 2 at the top and falls roughly as $\mathrm{e}^{-\tau}/\tau$ at depth.
- This is the working formula for thermal emission from an exoplanet without scattering: a black body in each layer plus the (much harder) opacities. Most retrieval codes evaluate exactly this sum.
- Written per unit $\log p$, the integrand $E_2(\tau)\, \mathrm{d}\tau/\mathrm{d}\log p$ is the **contribution function**: it shows which pressures a given wavelength actually probes.

Line formation in a toy atmosphere

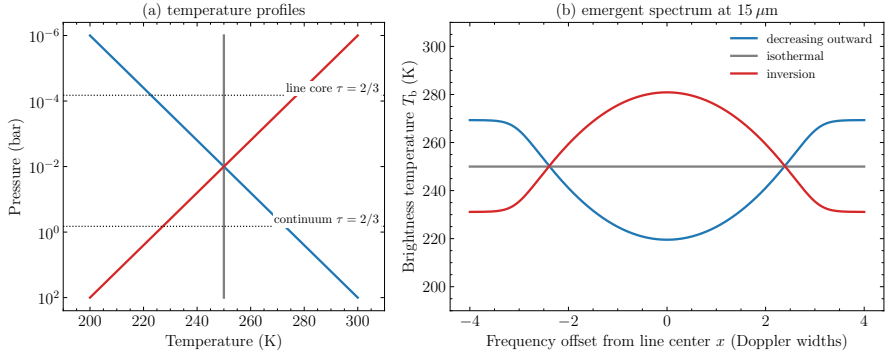


Figure 9: A line with a Gaussian opacity profile (peak 10^4 times the continuum) in three LTE atmospheres. Left: the temperature profiles and the pressures where $\tau = 2/3$ in the continuum and at line center. Right: the emergent flux as a brightness temperature. Computed with $F = 2\pi \int B E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$; script `make_line_formation.py`.

The continuum reports T near 0.7 bar, the line core T near 10^{-4} bar; the isothermal case is featureless despite the huge opacity contrast.

Why a band has a floor: the minimum flux

A molecular absorption band is made of many closely spaced lines (Chapter 8 of Seager), so from here on we speak of an absorption *band*. Take an atmosphere whose temperature decreases outward.

- **The question:** what is the *minimum* flux in the band — and why is it not zero?
- There is always some radiation, however little, emerging at every wavelength.
- In LTE the size of the emergent flux depends on the temperature of the altitude where that flux *originates*.

So the question becomes: from which altitude does the flux at each wavelength come? The answer is set by the opacity, which varies strongly with wavelength — next slide. In an LTE atmosphere the flux at any wavelength is never zero; it is the Planck function of whatever layer that wavelength "sees".

Maximum and minimum flux: Earth's 15 μm CO_2 band

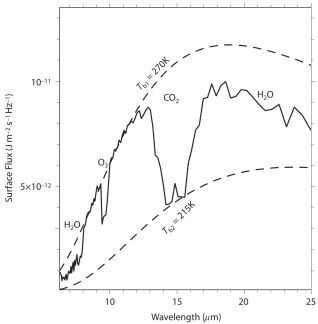


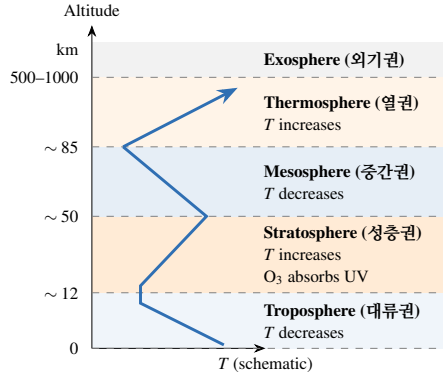
Figure 10: Earth's mid-infrared spectrum with black bodies at $T_{b1} = 270$ K and $T_{b2} = 215$ K. Reproduced from Seager (2010), Fig. 3.4.

- At a **low-opacity** wavelength the radiation can travel essentially unimpeded, so it emerges from *deep*, warm layers: this is the **maximum** flux, set by the surface or deep temperature.
- At a **high-opacity** wavelength the radiation is reprocessed in every layer on its way out; what finally escapes must have originated near the *top*, where it is cold: this is the **minimum** flux, set by the coldest layer the band can reach.
- Because opacity depends on wavelength, the spectrum *samples different temperatures* — in LTE, flux of different magnitude.
- **Earth:** between the bands the spectrum follows the warm $T_{b1} = 270$ K curve; the 15 μm CO_2 band bottoms out on the cold $T_{b2} = 215$ K curve, not at zero.
- The band depth therefore measures the *temperature range* between the continuum and band-center photospheres, not the amount of CO_2 as such. Brightness temperatures are a direct, if coarse, thermometer at different heights.

Reading Earth's temperature profile from its spectrum

- The deep O_3 (9.6 μm) and CO_2 (15 μm) absorption bands tell us that temperature decreases with height where they form: the **troposphere**, which holds most of the mass.
- At the very center of both bands there is a **narrow emission spike**. The core forms highest, so somewhere above the troposphere the temperature must rise again: the **stratospheric inversion**, heated by ozone absorbing solar UV.
- Why is the spike narrow? The opacity of a band scales with the number density of absorbers. Only at the band center is the opacity large enough for $\tau \approx 1$ to be reached in the low-density stratosphere; a little off-center, $\tau \approx 1$ drops back into the troposphere.
- The narrowness of the emission core tells us that the emission features are forming in regions of low density.

The width of an emission core thus encodes how high, and in how thin a layer, the inversion sits.



Approximate boundaries; height not to scale.
Tropopause height varies with latitude and season.
Source: UCAR Center for Science Education

Thermal inversions in hot Jupiters?

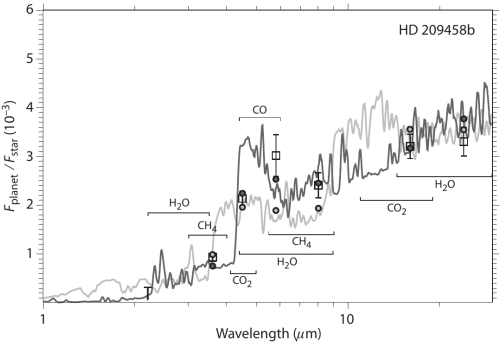


Figure 11: Historical interpretation of HD 209458b: Spitzer photometry (squares with error bars). Assuming H_2O and CO, the emission-only model (black) fits better than the absorption-only model (gray). Seager (2010), p. 112, Fig. 6.6; the inversion inference was revised by later observations.

- **Seager (2010):** secondary eclipses probe the permanent dayside of tidally locked hot Jupiters. Spitzer data suggested H_2O emission in some planets, absorption in others.
- **Earth:** weak emission cores within deep absorption bands indicate an inversion *above* the layers forming most of the absorption.
- **Initial HD 209458b interpretation:** strong emission without discernible absorption suggested an **on-average deep dayside inversion**, reaching the main feature-forming layers.
- **Update — HD 209458b:** HST/WFC3 resolved H_2O *absorption* (6.2σ). Together with Spitzer data, it supports **no inversion** over the probed $1–10^{-3}$ bar range (Line et al. 2016).
- **Update — WASP-121b:** resolved H_2O *emission* supports a stratospheric inversion (5σ; Evans et al. 2017).

Pure transmission: Lambert's law or Beer's law

In transmission there is neither emission nor scattering *into* the beam: $\varepsilon = 0$, hence $S = 0$, and

$$\mu \frac{\mathrm{d}I}{\mathrm{d}z} = -\kappa I \quad \Longrightarrow \quad I(z_f) = I(z_i) \exp\left[-\frac{1}{\mu}\int_{z_i}^{z_f} \kappa \, \mathrm{d}z'\right], \qquad I(\tau_\nu) = I(0)\, \mathrm{e}^{-\tau_\nu/\mu}.$$

In a transit the relevant quantity is the ratio of transmitted to incident stellar light along one ray, the stellar intensity cancelling:

$$\frac{F_{\mathrm{S},p,\mathrm{trans}}(\nu)}{F_{\mathrm{S},*,\mathrm{trans}}(\nu)} = \mathrm{e}^{-\tau_\nu},$$

where τ_ν is now the optical depth *along the chord* through the limb (Seager Eq. 6.21).

- This is one ray. The observed transit depth sums rays at all impact parameters through the annulus, each with its own chord optical depth (Lecture 3: $\tau_{\mathrm{chord}} \approx \sqrt{2\pi R_p/H}$ times the vertical one).
- Pressure broadening matters at depth, which is why the sum over altitudes cannot be skipped in a real calculation.

A transmission spectrum: HD 189733b

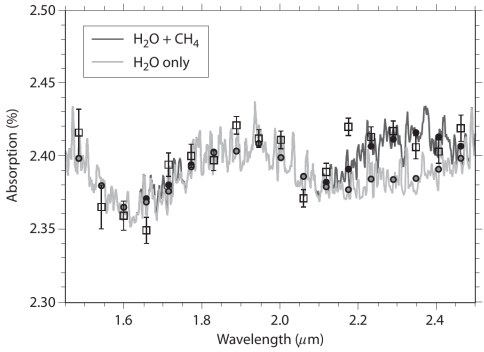


Figure 12: HST/NICMOS transmission spectrophotometry of HD 189733b (open squares with error bars). Curves: H_2O only (gray) or $H_2O + CH_4$ (black). Open circles and filled circles are, respectively, the H_2O-only and $H_2O + CH_4$ model predictions averaged over the observational wavelength bins, accounting for the instrument response. Seager (2010), Fig. 6.7.

- The signal is $\sim 5 \times 10^{-4}$ of the stellar flux, as estimated in Chapter 3: the area of an annulus about $5H$ tall relative to the stellar disk.
- Atoms and molecules detected in transmission in hot Jupiters include Na, K, H_2O, CO, CO_2 and (with JWST) SO_2.
- This particular CH_4 detection was later disputed: the NICMOS systematics had been underestimated (Gibson et al. 2011). A useful reminder that a feature is only as good as the instrument model behind it.

Seager's exercise 3(c) asks when thermal emission from the planet would matter in transmission: when the planet's own emission along the chord is not negligible compared with the transmitted starlight, i.e. for very hot planets observed in the infrared.

Thermal emission: the formal solution with $S = B$

At infrared wavelengths molecular absorption dominates and scattering can be neglected: $\kappa = \alpha_{\rm abs}$ (absorption only), and in LTE $S_{\rm therm} = \varepsilon/\kappa = B(\tau_\nu, \nu)$. The equation $\mu\,\mathrm{d}I/\mathrm{d}\tau_\nu = I - B$ has the solution

$$I(0,\mu,\nu) = \int_0^\infty B(\tau_\nu,\nu)\,\mathrm{e}^{-\tau_\nu/\mu}\,\frac{\mathrm{d}\tau_\nu}{\mu}, \qquad \boxed{F(0,\nu) = 2\pi\int_0^1\int_0^\infty B\,\mathrm{e}^{-\tau_\nu/\mu}\,\mathrm{d}\tau_\nu\,\mathrm{d}\mu}.$$

- "Powerfully simple": each layer is a black body; the only hard ingredient is the opacity, which fixes $\tau_\nu(z)$.
- Exact in LTE without scattering, given $T(z)$. This makes it the workhorse for secondary-eclipse and directly imaged spectra.
- Thermal emission peaks near the Planck maximum: ~ 0.8–$1\,\mu$m for 2000–3000 K ultra-hot Jupiters, $\sim 15\,\mu$m for Earth, longer for colder planets.
- Next slide shows data for one example, the Spitzer and Hubble Space Telescope observations of HD 189733b.

An emission spectrum: HD 189733b

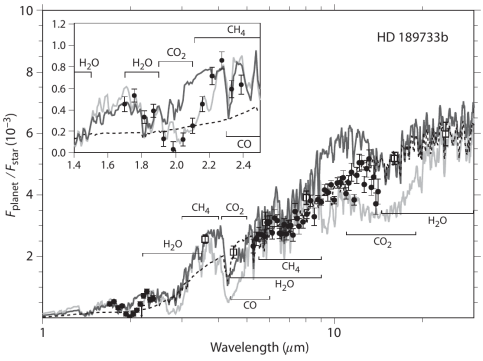


Figure 13: Secondary-eclipse data for HD 189733b: Spitzer photometry (squares), Spitzer/IRS (circles), HST/NICMOS (inset), with best-fit models. Seager (2010), Fig. 6.8.

- The planet/star contrast reaches $\sim 5 \times 10^{-3}$ in the mid-infrared, ten times the transmission signal.
- Order of magnitude (Chapter 3, Rayleigh–Jeans limit):

$$\frac{F_p}{F_*} \approx \frac{T_{\mathrm{b},p}}{T_{\mathrm{b},*}} \left(\frac{R_p}{R_*}\right)^2 .$$

  With $R_p/R_* = 0.155$ and $T_{\mathrm{b},*} \approx 5000\,\mathrm{K}$, a contrast of 3.4×10^{-3} at $8\,\mu\mathrm{m}$ implies $T_{\mathrm{b},p} \approx 700\,\mathrm{K}$; the full Planck ratio gives $\approx 1300\,\mathrm{K}$, close to the measured dayside value.
- Absorption features of H_2O and CO/CO_2: temperature decreasing with height on this dayside.

A constant source function

We can solve the RT equation under the assumption of a constant source function with altitude. Take S independent of depth between an upper level ($\tau_{\min}$) and a lower one ($\tau_{\max}$); along a vertical ray,

$$I(\tau_{\min}) = I(\tau_{\max})\,\mathrm{e}^{-(\tau_{\max}-\tau_{\min})} + S\left[1-\mathrm{e}^{-(\tau_{\max}-\tau_{\min})}\right].$$

The first term is what survives of the light entering from below, the second what the slab emits along the path.

- **Optically thick slab** (deep in the atmosphere), $\tau_{\max}-\tau_{\min} \gg 1$: $I = S$. The incoming light is forgotten; only local conditions matter, and in LTE $I = S = B$: a featureless continuum. *Lines do not form at large optical depth.*

$$I(0,\nu) = S(\nu).$$

- **Optically thin top layer**, $\tau_{\min} = 0$, $\tau_{\max} < 1$, so $\mathrm{e}^{-\tau} \approx 1-\tau$:

$$I(0,\nu) = I(\tau_{\max},\nu) - \tau_{\max}(\nu)\left[I(\tau_{\max},\nu) - S(\nu)\right].$$

  [Absorption Line Spectrum] The layer *removes* light if the background is brighter than its source function (i.e., $I(\tau_{\max},\nu) > S(\nu)$).

  [Emission Line Spectrum] The layer *adds* light if it is fainter (i.e., $I(\tau_{\max},\nu) < S(\nu)$), in proportion to its optical depth $\tau_{\max}(\nu)$.

Absorption or emission: the four cases

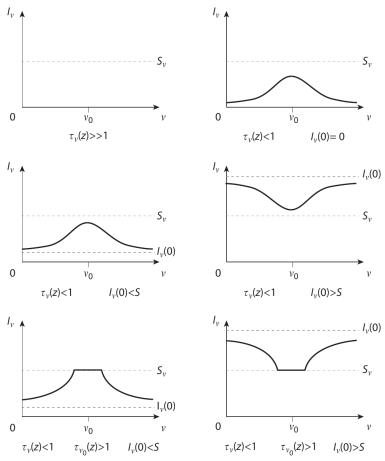


Figure 14: Line formation for a source function constant with depth and frequency. Seager (2010), Fig. 6.9, after Rutten.

Read the panels with $I(\tau_{\max}) \equiv I_\nu^0$, the light from behind, and $\tau_\nu(z)$ the layer's optical depth, largest at line center ν_0:

- **Top left** — thick at all ν: $I = S$, no line.
- **Top right** — thin, no backlight ($I_\nu^0 = 0$): emission line, strength $\propto \tau_\nu$.
- **Middle left** — thin, $I_\nu^0 < S$: emission line on top of the background.
- **Middle right** — thin, $I_\nu^0 > S$: absorption line.
- **Bottom** — thick at line center only ($\tau_{\nu_0} > 1$): the core *saturates* at $I = S$, giving a flat-topped emission or flat-bottomed absorption line.

Absorption lines require light from behind that is *brighter* than the layer's own source function — the same statement as "cool gas in front of a hot body".

A linear source function: Eddington–Barbier approximation

Let $S(\tau_\nu) = a_\nu + b_\nu\,\tau_\nu$ (independent of μ). The formal solution gives exactly

$$I_\nu(0,\mu) = a_\nu + b_\nu\mu, \qquad F_\nu(0) = 2\pi\int_0^1 (a_\nu + b_\nu\mu)\,\mu\,\mathrm{d}\mu = \pi\left(a_\nu + \tfrac{2}{3}b_\nu\right). \qquad \boxed{F_\nu(0) = \pi S(\tau_\nu = 2/3).}$$

- The emergent **flux** is characteristic of $\tau_\nu \approx 2/3$; the **intensity** along the vertical ray, $I(0,1) = S(\tau_\nu = 1)$. This is the Eddington–Barbier relation.[2]
- It does not require the absence of scattering, only a source function that is (close to) linear in τ.
- Because τ_ν depends on frequency, so does the physical height of $\tau_\nu = 2/3$: this is the mathematical form of **"the line core forms higher (outer layer)"**.
- The same $I = a_\nu + b_\nu\mu$ gives $\boxed{K(\tau_\nu,\nu) = J(\tau_\nu,\nu)/3}$ for the second and zeroth moments — the **Eddington approximation**, which reappears as the third Eddington coefficient of the two-stream method.

[2] Seager writes $F(0,\nu) = S(\tau_\nu = 2/3)$ and $F(0,\nu) = S(\tau_\nu = 1)$ (Eqs. 6.32–6.33); the first lacks the factor π, and the second is the vertical intensity rather than a flux.

A Scattering Atmosphere: Why scattering makes everything harder

Absorption and thermal emission tie the radiation to the *local* temperature. Scattering does not:

- A scattered photon can travel far before it is absorbed. The long-distance photon travel by scattering therefore **couples distant parts of the atmosphere**, and even the open boundary at the top influences the field deep inside.
- The source function now contains the radiation field itself,

$$S_{\text{scat}}(\tau,\mu,\phi) = \frac{\omega_0}{4\pi}\int_{4\pi} P(\mu,\phi;\mu',\phi')\,I(\tau,\mu',\phi')\,\mathrm{d}\Omega',$$

  so the RT equation becomes an **integro-differential equation**: I appears on both sides, once differentiated and once integrated over all directions.
- Few analytic solutions exist; classical texts spend many pages even on isotropic scattering (Chandrasekhar 1960; Sobolev 1975).

Here ω_0 ($\tilde{\omega}$ in Seager) is the single-scattering albedo and P the single-scattering phase function, normalized as $\int_{4\pi} P\,\mathrm{d}\Omega/4\pi = 1$ (Lecture 3).

Direct beam and diffuse field

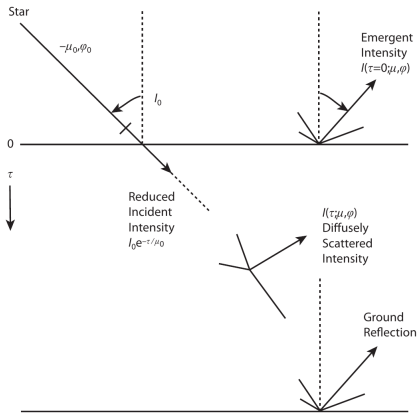


Figure 15: The attenuated direct beam from the star and the diffusely scattered intensity. Seager (2010), Fig. 6.10, after Chamberlain & Hunten.

Starlight arrives as a parallel beam from direction (μ_0, ϕ_0):

$$I_*(0,\mu,\phi) = I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

The incident flux follows from angular integration (Seager, p. 120, Eq. 6.37):

$$\begin{aligned} F_*(0,\mu,\nu) &= \int_0^{2\pi}\int_{-1}^{0} \mu I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0)\,\mathrm{d}\mu\,\mathrm{d}\phi \\ &= \mu_0 I_0. \end{aligned}$$

In this signed-flux equation, incoming $\mu_0 < 0$; the incident-flux magnitude is $|\mu_0| I_0$. For attenuation below, μ_0 denotes the positive magnitude.

Unscattered, the beam is attenuated as

$$I_{\mathrm{dir}}(\tau) = I_0\,\mathrm{e}^{-\tau/\mu_0}\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

It is convenient to split the field into this **direct beam**, known analytically, and the **diffuse field** I of photons scattered at least once, which is what we must solve for.

Scattering RT: an angularly coupled problem (6.4.3.2)

Let τ_ν increase inward and $\mu > 0$ denote outgoing rays. For coherent scattering, the source receives photons from all directions:

$$\boxed{\mu \frac{\mathrm{d}I_\nu(\tau_\nu,\Omega)}{\mathrm{d}\tau_\nu} = I_\nu(\tau_\nu,\Omega) - S_\nu(\tau_\nu,\Omega)}$$

$$S_{\mathrm{scat},\nu}(\tau_\nu,\Omega) = \frac{\omega_0}{4\pi}\int_{4\pi} P_\nu(\Omega;\Omega')I_\nu(\tau_\nu,\Omega')\,\mathrm{d}\Omega'.$$

- $\Omega = (\mu,\phi)$, $\mathrm{d}\Omega' = \mathrm{d}\mu'\,\mathrm{d}\phi'$; P is normalized by $\int P\,\mathrm{d}\Omega/4\pi = 1$. Pure scattering has $\omega_0 = 1$.
- Scattering removes photons from one ray and adds photons from other rays. Unlike an LTE thermal source $B_\nu(T)$, the scattering source depends on the *unknown radiation field*.
- The equation is therefore **integro-differential**: every direction is coupled to the others (Seager Eqs. 6.38–6.40).

Notation correction: use $I - S$, as in Seager Eq. 6.1; the source is an integral over solid angle, without a flux factor μ'. The printed Eqs. 6.38–6.40 have inconsistent signs/angular measures.

Separate the direct beam from diffuse scattering (6.4.3.2)

Write $I_{\rm tot} = I_{\rm dir} + I_{\rm diff}$. Here $\mu_0 > 0$ is the *magnitude* of the incoming cosine, so the stellar beam travels in direction $\Omega_0 = (-\mu_0, \phi_0)$.

$$S_{{\rm single},\nu}(\tau_\nu, \Omega) = \frac{\omega_0 I_0}{4\pi}\, \mathrm{e}^{-\tau_\nu/\mu_0} P_\nu(\Omega; \Omega_0), \qquad \text{(Seager 6.43)}$$

$$S_{{\rm diff},\nu}(\tau_\nu, \Omega) = \frac{\omega_0}{4\pi} \int_{4\pi} P_\nu(\Omega; \Omega') I_{{\rm diff},\nu}(\tau_\nu, \Omega')\, \mathrm{d}\Omega', \qquad (6.44)$$

- The direct beam is known analytically. Its source term describes the **first scattering of stellar photons**; the diffuse term accounts for photons that scatter again.
- For an outgoing ray, a layer of total depth τ_b gives

$$\boxed{I_\nu(0, \mu) = I_\nu(\tau_b, \mu)\mathrm{e}^{-\tau_b/\mu} + \frac{1}{\mu}\int_0^{\tau_b} S_\nu(t, \mu)\mathrm{e}^{-t/\mu}\, \mathrm{d}t}$$

- This formal solution is still valid, but $S_{\rm diff}$ contains the unknown $I_{\rm diff}$: it must be solved self-consistently.

Seager Eqs. 6.41–6.42 rewritten with the required $1/\mu$ and an explicit lower-boundary term. For stellar scattering alone, $S = S_{\rm single} + S_{\rm diff}$; thermal emission adds $(1 - \omega_0) B_\nu$.

Single scattering in a thin layer (6.4.3.3)

Optically thin cirrus or aerosol layers high in the atmosphere can be dominated by one scattering of the incident stellar beam. Set $S = S_{\text{single}}$ and neglect illumination from below:

$$I_\nu(0,\mu) = \frac{\omega_0 I_0 P_\nu}{4\pi\mu} \int_0^{\tau_b} \mathrm{e}^{-t/\mu_0} \mathrm{e}^{-t/\mu}\, \mathrm{d}t = \frac{\omega_0 \mu_0 I_0 P_\nu}{4\pi(\mu+\mu_0)} \left[1 - \mathrm{e}^{-\tau_b(1/\mu+1/\mu_0)}\right].$$

The exponentials account for attenuation **before** and **after** scattering (Seager Eqs. 6.45–6.46). For $\tau_b/\mu \ll 1$ and $\tau_b/\mu_0 \ll 1$,

$$\boxed{R(\mu,\phi;\mu_0,\phi_0) \equiv \frac{\pi I_\nu(0,\mu,\phi)}{\mu_0 I_0} \simeq \frac{\omega_0 \tau_b}{4\mu\mu_0} P_\nu(\mu,\phi;-\mu_0,\phi_0)} \qquad (6.47)$$

- R is Seager's **dimensionless bidirectional reflectance** (BRDF): a Lambert-normalized reflected intensity. On Earth, the BRDF can be determined from satellite measurements.
- At fixed geometry, albedo and phase function, $R \propto \tau_b$. No isotropic-scattering or two-stream assumption is needed.
- A small *vertical* depth is insufficient near grazing rays: the relevant incoming and outgoing *slant* depths must be small.

Eq. 6.45 correction: the incoming attenuation is e^{-t/μ_0}, not $\mathrm{e}^{-t}\mu_0$. The finite-depth expression is exact for the single-scattering term, not for unrestricted multiple scattering.

Multiple scattering and Earth's spectrum (6.4.3.4)

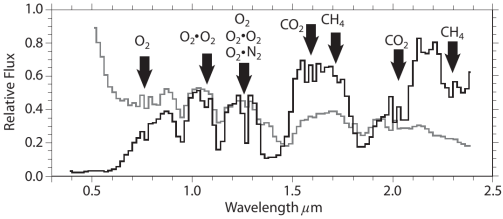


Figure 16: Earth as an exoplanet. Both spectra were observed by telescopes on Earth, using the Moon as a reflector. Gray (earthshine): Sun → Earth (scattering/reflection) → Moon → telescope on Earth. Black (lunar eclipse): Sun → Earth's atmospheric limb (transmission/refraction) → Moon → telescope on Earth. The long slant paths in transmission and the cloud/surface-weighted paths in scattered light produce different band strengths. Labels identify O_2, O_2–O_2 and O_2–N_2 collision-induced absorption, CO_2 and CH_4. Relative fluxes emphasize spectral shapes, rather than absolute planet/star contrasts. Seager (2010), Fig. 6.11; Pallé et al. (2009).

- Once repeated scattering matters, the source and radiation field are coupled; a simple single-scattering solution is no longer sufficient.
- **Adding–doubling:** calculate reflection and transmission for a thin homogeneous layer. Combine layers, including repeated reflections between them; doubling builds a thick slab efficiently.
- Different homogeneous layers can be added to represent a **vertically inhomogeneous** atmosphere.
- Two hot Jupiters had visible geometric-albedo upper limits below 0.15, compared with Jupiter's ~ 0.5; this helped explain the difficulty of scattered-light detection.
- Earth's scattered and transmitted spectra probe different paths and weight clouds and gases differently. Their molecular-band strengths therefore differ (left figure).

Diffusion deep in the atmosphere (6.4.4)

At $\tau_\nu \gg 1$, temperature and opacity vary little over a photon mean free path. In the thermalized, LTE regime, $S_\nu \simeq B_\nu(T)$. Expand locally:

$$B_\nu(t) \simeq B_\nu(\tau_\nu) + (t - \tau_\nu)\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

With $\mu\,\mathrm{d}I_\nu/\mathrm{d}\tau_\nu = I_\nu - B_\nu$, the leading solution is

$$\boxed{I_\nu(\tau_\nu, \mu) \simeq B_\nu(\tau_\nu) + \mu\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}} \qquad \text{(6.49, consistent sign)}$$

- The leading term is isotropic: $I_\nu \simeq S_\nu \simeq B_\nu$. A small angular anisotropy carries the net flux.
- Integrating over *both* hemispheres cancels the isotropic term:

$$\boxed{F_\nu = 2\pi\int_{-1}^{1}\mu I_\nu\,\mathrm{d}\mu \simeq \frac{4\pi}{3}\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}}$$

- This is a **deep-interior approximation**, not a solution for the escaping radiation at $\tau_\nu \lesssim 1$.

With optical depth increasing inward, Eq. 6.49 needs a plus sign to agree with the RT equation and Eq. 6.51. Large scattering depth alone does not guarantee thermalization.

Radiative diffusion: opacity and the temperature gradient

Let χ_ν be the extinction per unit length, $\mathrm{d}\tau_\nu = -\chi_\nu\,\mathrm{d}z$, with z increasing outward. For isotropic scattering (or using the appropriate transport opacity),

$$F_\nu \simeq -\frac{4\pi}{3\chi_\nu}\frac{\partial B_\nu}{\partial T}\frac{\mathrm{d}T}{\mathrm{d}z}.$$

The frequency-integrated flux uses the **Rosseland mean opacity**:

$$\frac{1}{\chi_R} = \frac{\displaystyle\int_0^\infty \chi_\nu^{-1}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}, \qquad \chi_R = \rho\,\kappa_R \quad \text{if } \kappa_R \text{ is per unit mass.}$$

Since $\int B_\nu\,\mathrm{d}\nu = \sigma T^4/\pi$,

$$\boxed{F(z) = -\frac{16\sigma T^3}{3\chi_R}\frac{\mathrm{d}T}{\mathrm{d}z}}.$$

- Flux travels **opposite to the temperature gradient**, like heat conduction: $F = -k_{\mathrm{rad}}\,\mathrm{d}T/\mathrm{d}z$, $k_{\mathrm{rad}} = 16\sigma T^3/(3\chi_R)$.
- The harmonic mean emphasizes transparent spectral windows: photons diffuse most easily through low-opacity channels.

Notation clarification: the monochromatic expression uses χ_ν; the Rosseland mean enters after frequency integration. Seager writes the per-length mean as κ_R.

Radiative transfer in cloudy atmospheres

The same RT equations apply to clouds. The numerical difficulty is the **sharp opacity change at an optically thick cloud boundary**. Seager (p. 125) summarizes several practical treatments:

1. **Cloud top as a boundary:** place the effective atmospheric surface at the top of an opaque cloud and prescribe its emerging radiation, instead of solving through the cloud interior.
2. **Partial cloud coverage:** combine cloudy and clear columns, possibly with different cloud-top heights:

$$\boxed{I_{\text{disk}} \simeq f_{\text{cl}} I_{\text{cloudy}} + (1 - f_{\text{cl}}) I_{\text{clear}}}.$$

   Here f_{cl} is the projected-area cloud fraction; the columns are assumed radiatively independent.
3. **Smooth the boundary:** replace an abrupt cloud edge with an exponential density decrease for numerical convenience.
4. **Approximate cloud scattering analytically:** prescribe the radiation emerging from a cloud layer without solving its full internal transfer.

For realistic, nonuniform **3D cloud structures**, **Monte Carlo RT** approach is the most accurate way to deal with scattering from clouds.

Monte Carlo RT: follow photon packets

Traditional RT solves for the intensity field; Monte Carlo RT samples **individual photon-packet histories**. Their ensemble estimates the same radiation field.

- A packet represents many photons at frequency ν, with a position, direction and energy; polarization can also be tracked.
- Draw the next interaction distance, absorption or scattering, and the new direction from the appropriate probability laws. Follow each packet until it is **absorbed or escapes**.
- **Strength:** irregular geometry and spatially varying properties, including clumpy 3D clouds, can be treated directly.
- **Cost:** many histories are needed to reduce sampling noise; optically thick regions require many scattering events. Seager discusses simulations with millions of packets or more.

Scope here: time-independent, monochromatic, coherent scattering. The method can be applied to 3D media, thermal emission and temperature-structure calculations.

Sampling a free path: inverse cumulative probability

Let $p(x)$ be a normalized probability density on $[a,b]$. Draw ζ uniformly in $(0,1)$ and invert its cumulative distribution:

$$\boxed{\zeta = \int_a^{x_0} p(x)\,\mathrm{d}x, \qquad \int_a^b p(x)\,\mathrm{d}x = 1}.$$

The probability of no interaction after a path optical depth t is e^{-t}. Thus the next-interaction probability density is

$$p(t)\,\mathrm{d}t = \mathrm{e}^{-t}\,\mathrm{d}t, \qquad t \geq 0.$$

$$\zeta = 1 - \mathrm{e}^{-t} \quad \Longrightarrow \quad \boxed{t = -\ln(1-\zeta)}.$$

- t is optical depth *along the packet's ray*, not vertical depth. Convert it into a distance s using the local extinction: $t = \int_0^s \chi_v(\mathbf{r} + s'\hat{\mathbf{n}})\,\mathrm{d}s'$.
- In a homogeneous medium, $s = t/\chi_v$. If a boundary is closer than the sampled interaction, the packet reaches that boundary first.

The photon-packet algorithm

1. **Initialize.** Launch a packet from the stellar source; set its entry position, direction, frequency and energy.
2. **Propagate.** Sample $t = -\ln(1-\zeta_1)$ and find the corresponding path length. If the packet exits first, record it; otherwise move to the interaction point.
3. **Choose the interaction.** With a new random number, scatter if $\zeta_2 < \omega_0$ and absorb otherwise:

$$\boxed{\Pr(\text{scatter}) = \omega_0, \qquad \Pr(\text{absorb}) = 1-\omega_0}.$$

4. **Scatter and repeat.** Draw a new outgoing direction from $p(\Omega \mid \Omega') = P(\Omega;\Omega')/(4\pi)$; update polarization if included, then return to step 2. Coherent scattering preserves ν.
5. **Absorb or escape.** Absorption deposits the packet energy and ends its contribution to this monochromatic scattering calculation. Escaping packets contribute to the emergent radiation.

Absorption does not destroy energy: re-emission at thermal wavelengths is outside this scattering-only example. A surface encounter requires an appropriate absorption/reflection boundary condition.

From escaping packets to observables and errors

Bin escaped packets by direction, exit location and frequency. Their energy, normalized to the injected luminosity and bin geometry, gives the emergent flux; directional bins describe phase and inclination dependence (and polarization, if tracked).

- **Equal-energy packets:** if each packet carries energy $E_{\rm pkt}$ over a normalization time Δt, then $L_{\rm bin} = N_{\rm bin} E_{\rm pkt}/\Delta t$. Conversion to intensity or observer flux must include the relevant area, solid angle and distance.
- **Sampling error:** Seager uses Poisson counting errors. For independent, equal-weight packets and well-populated bins,

$$\boxed{\sigma_N \simeq \sqrt{N_{\rm bin}}, \qquad \frac{\sigma_F}{F} \simeq \frac{1}{\sqrt{N_{\rm bin}}}}.$$

  Halving the statistical error requires roughly four times as many packets in that bin.
- **Numerical check:** in a steady scattering-only run, escaped energy plus absorbed energy should match injected energy.
- Monte Carlo naturally handles complex clouds, but noise and long random walks make optically thick calculations expensive. Weighted-packet methods require their own variance estimates.

Take-home messages

Albedos. A perfect Lambert surface has $p = 1$ by energy conservation; a Lambert sphere has $A_g = 2/3$, $q = 3/2$, $A_S = 1$. A_g can exceed 1; A_B is A_S weighted by the stellar spectrum and enters the energy balance.

Flux ratios. Thermal: $F_p/F_* = (R_p/R_*)^2 B(T_p)/B(T_*)$, $\to (R_p/R_*)^2 T_p/T_*$ in the Rayleigh–Jeans limit. Scattered light at full phase: $F_p/F_* = A_g(R_p/a)^2$. Transmission: the annulus of a few scale heights, $\sim 2R_pH/R_*^2$.

Phase curves. $\Phi_\alpha = \Psi_\alpha/\Psi_{\alpha=0}$; a Lambert sphere has $\Phi_\alpha = [\sin\alpha + (\pi - \alpha)\cos\alpha]/\pi$. Infrared phase curves map the day–night temperature contrast.

Line formation. Spectral features measure temperature *contrasts*: an isothermal LTE atmosphere emits a featureless black body. Opaque wavelengths form high, transparent ones deep; temperature falling outward gives absorption, an inversion gives emission. Emergent flux $\approx \pi S(\tau_\nu = 2/3)$.

Solutions. Transmission: Beer's law along the chord. Thermal emission: $F = 2\pi \int B E_2 \, \mathrm{d}\tau$, exact without scattering. Scattering couples directions; thin layers give $R \propto \omega_0 \tau P/(4\mu\mu_0)$. Deep layers obey $F = -16\sigma T^3(\mathrm{d}T/\mathrm{d}z)/(3\chi_R)$; clouds require boundary treatments or explicit scattering calculations.

Monte Carlo. Sample photon-packet paths and interactions; escaped packets give observables with counting noise $\propto N^{-1/2}$.

Next. Numerical methods and retrieval, and the two-stream approximation (Lecture 6).

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